Case 12 Arch Coal Laying Off 750 Workers in Appalachia
By Bruce Schreiner on June 21, 2012
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Forty percent of U.S. energy will come from coal this year

One of the world's largest coal producers said Thursday it would lay off about 750 workers in the Kentucky, Virginia and West Virginia coalfields, the latest setback for an industry struggling to sustain market share as utilities switch to cleaner and cheaper alternatives to generate electricity. The bulk of the cuts by Arch Coal Inc., almost 600, are in Kentucky. The disappearance of high-paying mining work heightened anxiety in hardscrabble Kentucky towns where officials worried declining demand for coal would result in leaner budgets and more people on unemployment rolls.

"This is just a start, I think," said Dennis Ray Noble, the judge-executive of Perry County, which he estimated has lost about 30 percent of its mining jobs in the last year and the jobless rate is 12.4 percent.

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The St. Louis-based company said its subsidiaries would close three higher-cost mining complexes and associated preparation plants: two in Kentucky and one in West Virginia. It will temporarily idle another complex in Kentucky and curtail production at other facilities in the three states. The company accounted for 16 percent of the coal production in the U.S. in 2010. The layoffs come amid forecasts that the share of U.S. electricity coming from coal will fall below 40 percent for the year — the lowest level since the government began collecting data in 1949. Four years ago, it was 50 percent. By the end of this decade, it is likely to be near 30 percent.

"Current market pressures and a challenging regulatory environment have pushed coal consumption in the United States to a 20-year low," Arch President and CEO John W. Eaves said in a news release announcing the layoffs. Eaves said the company regrets the impact on its employees and their families and communities, but said the decision was necessary "to weather the current downturn and to position the company for long-term success."

Arch said the reductions would trim its thermal coal production by more than 3 million tons annually. It said it continues to expect thermal coal sales volume in the range of 128 million to 134 million tons for 2012. Meanwhile, the company said it is putting more emphasis on higher-margin metallurgical coal operations in the region.

The job losses come as utilities increasingly switch from coal to natural gas, which has become cheaper as supplies grow. Natural gas' advantages over coal include producing fewer emissions of toxic chemicals and gases that contribute to climate change. From Kentucky courthouses to Congress, officials blamed tougher environmental rules for coal's woes and the resulting job losses.

"I think they saw the writing on the wall that it's getting too difficult to operate at a profit," Knott County Judge-Executive Randy Thompson said of the Arch layoffs.
He said about 250 jobs will be lost from his county, which had a 12.5 percent jobless rate in May. Meanwhile, coal severance tax revenues are dwindling, putting more strain on the local budget to keep up with demand for services.

Republican U.S. Rep. Harold "Hal" Rogers, who represents the eastern Kentucky coalfields, said a mild winter, low demand for thermal coal and cheap natural gas prices have factored into the coal industry's struggles. But he said President Barack Obama's administration is also culpable because of deadlocked mining permits, rule changes and "crippling" regulations on power plants. "The Obama administration has continually kicked the industry while it's down and shown total disregard for the people of our region," Rogers said.

Power plants that burn coal produce more than 90 times as much sulfur dioxide, five times as much nitrogen oxide and twice as much carbon dioxide as those that run on natural gas, according to the Government Accountability Office, the regulatory arm of Congress.

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Sulfur dioxide causes acid rain; nitrogen oxides cause smog; and carbon dioxide is a so-called greenhouse gas that traps heat in the atmosphere.

A pair of clean air rules enacted by the Environmental Protection Agency over the past year tightens limits on power-plant emissions of sulfur dioxide and nitrogen dioxide, and place new limits on mercury, a poison found in coal. This will force between 32 and 68 of the dirtiest and oldest coal plants in the country to close over the next three years as the rules go into effect, according to an AP survey of power plant operators conducted late last year.

In what could be an even bigger environmental blow to coal, the EPA in March issued guidelines that could limit greenhouse gas emissions from new power plants. Robert Ukeiley, an environmental lawyer in Berea, Ky., said it's ridiculous to blame environmental regulations for the coal industry's struggles. He said those rules are "completely reasonable," have been decades in the making and are a "moral no-brainer."

"The fact is that cheap coal in central Appalachia has been mined out," he said. "That's just a fiscal fact. There is no politician who can change that or scapegoat anybody." Remaining coal supplies tend to be more expensive to extract and aren't as profitable, he said.

"Central Appalachia would be experiencing almost as much decline in coal mining even if we had the most anti-environmental president of our history in place," Ukeiley said. The EPA said in a statement that coal is still expected to generate more of the country's electricity than any other fuel source. It pointed to low natural gas prices, low electricity demand and rising coal prices as factors in reducing demand.

"Market conditions in the power sector are driving business decisions that are completely independent from these long-overdue toxic pollution standards, which power plants do not need to meet until 2016," the agency said in the statement.

Analysis

1. Identify what “miner layoff” influences and what influences it. Quantify the variables identified.
2. Develop a reinforcing loop.
3. Create two (2) balancing loops on top of the existing loop. Evaluate its significance.

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